AARYAV GANGRADE

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Education

Vellore Institute of Technology, Chennai **2022 – 2026**
Bachelor of Technology in Computer Science and Engineering CGPA: 7.61/10

Technical Skills

Languages: Python, SQL, Java, C++, JavaScript
Data Engineering & Cloud: PySpark, Spark SQL, Delta Lake, AWS S3, AWS IAM, Pandas, Parquet, ETL, Data Modeling
Backend & AI: FastAPI, REST APIs, FAISS, BM25, NetworkX, RAG, LLM Agents
Tools: Git, Linux, Databricks Notebooks, Streamlit

Projects

RetailLake – AWS-Backed Data Lakehouse
GitHub | Live Demo
Python, PySpark, Delta Lake, AWS S3, SQL

- Built a Bronze–Silver–Gold pipeline using **PySpark** to clean and organize **245K+ retail records** into **7 Delta Lake tables**, checking data types, missing values, and duplicate IDs.
- Used **AWS S3** to store and verify data; added error checks, batch tracking, failure logs, and restricted **AWS IAM** access to improve reliability and avoid repeated work.
- Verified **R$13.22M** in delivered-item sales across reports, completed the full pipeline in **226 seconds**, and published an interactive sales dashboard.

Codebase Intelligence System – Agentic Graph-RAG
GitHub
Python, FAISS, BM25, NetworkX, Gemini API, Streamlit

- Built an AI assistant that answers questions about Python codebases using **AST parsing**, **FAISS/BM25 search**, and **NetworkX call graphs** to find relevant code.
- Added **Gemini** to rank search results and search further when needed. In earlier tests on **31,441 lines of code** across **26 files**, parsing took **356 ms** and retrieval about **340 ms**.
- Selected fewer than **700 relevant lines** per question from a 31,441-line codebase, reducing the code sent to the AI by at least **97.8%** in earlier tests.

NSE W/M Pattern Scanner
GitHub
Python, Pandas, yfinance, Parquet, Concurrent Processing

- Built a scanner for **2,600+ NSE stocks** to find W-shaped and M-shaped price patterns, using parallel data downloads and **Parquet caching** to reuse saved data.
- Backtested W/M strategies over 2 years, recording **+22.70%** (W, NSE equities) and **+11.36%** (M, Nifty 500) simulated returns before trading costs.

Experience

NZESSENTIALS Remote, New Zealand
Freelance Python Developer May 2025 – July 2025

- Built parts of a digital-signature platform that connects web and desktop applications to capture, store, and verify signatures.
- Developed **REST APIs** to share data between applications and retrieve and validate stored signatures.

Leadership

Linux Club, VIT Chennai Mar 2023 – Mar 2024
Operations Department Lead

- Led operations for a **150-member** student club, organized technical workshops and events, and secured **INR 100,000** in sponsorships.

Certifications

Supervised Machine Learning: Regression and Classification – DeepLearning.AI Verify

GEN AI Using IBM Watsonx – IBM Career Education Program